Lecture 26: The "Ford F-150" of Nuclear Energy (The Standard Commercial PWR)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: Standardization

While early test reactors explored exotic geometries, coolants, and breeder configurations, the commercial utility industry rapidly coalesced around a simpler, highly robust design.

- **The Winner:** The Pressurized Water Reactor (PWR), originally designed by Admiral Rickover for naval propulsion, was scaled up for civilian power by Westinghouse (and later Framatome/Areva).
- **Dominance:** Today, roughly 65% of the world's commercial power reactors are PWRs.
- **Key Philosophy:** Keep the water *liquid*. Do not allow bulk boiling in the core. Strictly separate the radioactive primary loop from the clean secondary steam cycle.

1 The Scale of the Machine

To put the neutronic and thermal math into perspective, we must first understand the physical scale of a standard commercial PWR (often referred to as the 1000 MWe class).

Despite generating enough energy to power a million homes, the actual nuclear core is surprisingly compact, which explains why our thermal constraints are so severe.

- **Thermal Power:** $\approx$ 3000 MWth (Total heat generated by fission in the core)
- **Electric Power:** $\approx$ 1000 MWe (Factoring in the $\approx$ 33% thermodynamic Carnot limit of the secondary steam cycle)
- **Core Diameter:** $\approx$ 3.5 meters (11.5 feet)
- **Active Core Height:** $\approx$ 4.0 meters (12 to 14 feet)
- **Fuel Assemblies:** $\approx$ 157 to 193 (Depending on a 3-loop or 4-loop Westinghouse design)

2 The Primary Loop Architecture

The PWR is defined by its "Two-Loop" system. The highly radioactive water remains permanently sealed inside the thick concrete containment building.

2.1 1. The Primary Circuit

Water at high pressure ($\approx$ 15.5 MPa / 2250 psia) circulates through the core, absorbing massive amounts of heat but remaining subcooled (no bulk boiling).

- T_{in} (Cold Leg) $\approx 290°$C
- T_{out} (Hot Leg) $\approx 325°$C

2.2 2. The "Lung" of the System: The Pressurizer

Because liquids are functionally incompressible, a solid water loop would experience catastrophic pressure spikes from even tiny temperature fluctuations. To absorb these surges, the PWR relies on the **Pressurizer**.

- **Location:** Connected via a surge line to one of the Hot Legs.
- **State:** It is the *only* place in the primary system where water deliberately exists at saturation ($T_{sat} \approx 345°$C). It contains a steam bubble at the top (cushion) and liquid at the bottom. Electric heaters boil water to increase pressure, while cold-water sprayers condense steam to lower pressure.

Safety Warning: The "Solid" Plant
If the steam bubble in the pressurizer collapses (the plant goes "solid"), the primary loop loses its compressibility. Because water is incompressible, minor temperature fluctuations can cause catastrophic pressure spikes. At Three Mile Island (1979), operators prematurely throttled safety injection because they feared "going solid," unaware that the high level in the pressurizer was actually being caused by steam voids in the core pushing water upward.

2.3 3. The Steam Generators (The Massive Heat Exchangers)

The hot primary water flows through thousands of inverted U-tubes inside the Steam Generator (SG).

- **Primary Side (Tube Side):** Hot, radioactive water flows inside the tubes.
- **Secondary Side (Shell Side):** Clean feedwater surrounds the outside of the tubes. Operating at a lower pressure ($\approx$ 1000 psia), this secondary water boils violently, sending clean steam out the top to the turbines.

2.4 4. The Reactor Coolant Pumps (RCPs)

To move thousands of gallons per second, PWRs use massive, vertical, single-stage centrifugal pumps.

- **The Engineering Challenge:** The pump motor sits in the dry containment atmosphere, but the impeller is inside the 2250 psia radioactive water. Developing a highly reliable, multi-stage, controlled-leakage shaft seal to prevent primary coolant from blowing out along the rotating shaft is one of the greatest mechanical engineering achievements of the PWR.

3 Reactor Vessel and Containment: The Defense in Depth

The PWR relies on multiple physical barriers to prevent the release of fission products. This "Defense in Depth" philosophy is personified by the two most massive components of the plant.

3.1 1. The Reactor Pressure Vessel (RPV)

The RPV is the heart of the primary system. It must withstand 15.5 MPa (2250 psi) at 325°C while being subjected to intense fast-neutron bombardment.

- **Material:** Low-alloy carbon steel (e.g., SA-508) roughly 20–25 cm (8–10 inches) thick.
- **Cladding:** The inner surface is "buttered" with a thin layer of stainless steel to prevent corrosion of the carbon steel.
- **Neutron Embrittlement:** Over decades, fast neutrons displace atoms in the steel lattice, making the metal brittle. This "Nil-Ductility Transition Temperature" shift is the primary limiting factor for extending the life of a nuclear plant beyond 60 years.

3.2 2. The Containment Building

The "final" barrier. This is the massive concrete dome visible from miles away.

- **Design Basis:** It must be able to withstand a "Large Break LOCA" (Loss of Coolant Accident)—essentially the instantaneous snapping of the largest primary pipe.
- **Construction:** 3–4 feet of steel-reinforced concrete lined with a leak-tight carbon steel plate.

4 Fuel Management and Burnup

Commercial utilities do not replace all the fuel at once. Instead, they treat the core like a rotating inventory.

4.1 1. The 18-Month Cycle

Every 18 to 24 months, the plant shuts down for refueling.

- **The 1/3 Core Rule:** Typically, only the "oldest" third of the fuel assemblies (those that have been in for 3 cycles) are removed and sent to the spent fuel pool.
- **Shuffling:** The remaining 2/3 are shuffled to new positions (often moving from the center to the periphery) to ensure an even power distribution.

4.2 2. Defining "Burnup"

We measure how much energy we extract from the fuel in units of **Megawatt-days per Metric Ton of Uranium (MWd/MTU)**.

- **Modern PWR Limit:** $\approx 45,000$ to $60,000$ MWd/MTU.
- **Efficiency:** High burnup is desirable for economics, but is limited by "cladding creep" and the buildup of gaseous fission products (Xe, Kr) inside the fuel rod, which increases internal pressure.

5 Typical Plant Power Balance

To understand the "Engineering Efficiency," we look at where the energy goes. A standard 4-loop Westinghouse plant might look like this:

Component	**Power Level**	**Notes**
Core Thermal Power (Q_{th})	3411 MWth	Total heat from fission/decay
RCP Heat Input	+14 MWth	Large pumps actually add heat!
Total Steam Heat	**3425 MWth**	Delivered to Steam Generators
Gross Electric Output (W_{gross})	1150 MWe	Generated by the Main Turbine
House Loads (Pumps/Fans)	-40 MWe	Electricity used to run the plant
Net Electric Output (W_{net})	**1110 MWe**	What actually goes to the grid

- **Net Thermal Efficiency:** $\eta = \frac{W_{net}}{Q_{th}} \approx 32.5\%$.
- **Heat Sink:** The remaining $\approx$ 2300 MW of heat is rejected to the environment (river, lake, or cooling tower).

6 Core Geometry: Driven by Thermal Limits

Before we look at neutronics, we must build the core grid. As we derived in Lecture 25, the size of the fuel rods is strictly dictated by heat transfer, not nuclear physics.

6.1 Recall: The R^2 Limit

Uranium Dioxide (UO_2) is the standard fuel because it is chemically stable and resists radiation swelling. However, it is a ceramic with very poor thermal conductivity ($k_f \approx 3.0$ W/m $\cdot$ K).

Recall the centerline temperature derivation for a solid cylinder with internal heat generation (q'''):

$$\Delta T_{pellet} = T_{center} - T_{surface} = \frac{q''' R^2}{4k_f} \tag{1}$$

Because the internal temperature drop scales with the **square of the radius (R^2)**, we cannot simply build "fat" fuel rods to hold more uranium. If we doubled the radius of a standard rod, the internal ΔT would quadruple, causing the center of the UO_2 pellet to melt ($\approx 2800°$C). Therefore, thermal conduction strictly limits the fuel to a "thin rod" geometry (roughly 1 cm in diameter).

7 The Core Physics: Mean Free Path and Moderation

Once the rod size is fixed by thermal limits ($\sim$ 1 cm), we tune the *spacing* (pitch) between the rods based on neutronics.

7.1 1. The Fast Neutron (2 MeV): Transparent Lattice

When a neutron is born from fission, it has high energy.

- **Physics:** The scattering Mean Free Path (λ_s) in water is $\approx$ **3** to **5** cm.

- **Geometry:** The Fuel Rod Pitch (center-to-center distance) is only **1.26** cm.
- **Result:** The fast neutron essentially "ignores" the physical lattice structure, traveling past 4 or 5 rods before undergoing significant collision.

7.2 2. The Thermal Neutron (0.025 eV): Optimizing the Gap

Once the neutron becomes thermal, its behavior changes drastically.

- **Physics:** The Mean Free Path in water drops to $\lambda_{thermal} \approx \mathbf{0.3}$ cm.
- **Geometry:** The actual physical water gap between the rods is roughly **0.31** cm.
- **Optimization (H/U Ratio):** This is no coincidence. The water gap isize is of the same order as the thermal mean free path, which helps neutrons return to fuel rather than being absorbed in water.

8 The Assembly Structure: The 17x17 Standard

Commercial reactors don't load individual pins; they load massive, highly engineered bundles. The industry standard (Westinghouse) uses a 17×17 square lattice.

- **Dimensions:**
  - Rod Diameter: 0.950 cm.
  - Clad Thickness: 0.057 cm (Zircaloy-4 or ZIRLO).
  - Array: 264 Fuel Rods, 24 Guide Thimbles, 1 Central Instrument Tube.
- **Guide Thimbles:** These are empty tubes replacing fuel rods in specific locations. They provide a clear structural channel for control rods or instruments to be inserted directly into the assembly.
- **Spacer Grids & Mixing Vanes:** To keep the 12-foot-long rods from bowing and touching each other, they are held by spring-loaded Zircaloy/Inconel grid straps. Furthermore, these grids contain small **mixing vanes** that induce swirl in the water, stripping away the boiling boundary layer to prevent DNB (as discussed in Lecture 25).

9 Control Strategy 1: The Mechanical "Spider" (RCCA)

Early reactors used large cruciform blades that caused severe "flux dipping" and localized power spikes. The modern PWR solves this with the **Rod Cluster Control Assembly (RCCA)**.

- **Design:** 24 thin absorber rods (made of Silver-Indium-Cadmium) are attached to a common top hub (the "spider"). These fingers slide perfectly down into the 24 Guide Thimbles of the fuel assembly.
- **Advantage:** Distributing the absorber material evenly throughout the assembly keeps the neutronic power profile much flatter.
- **Safety Mechanism:** They are held vertically above the core by electromagnetic coils. In the event of a power failure, the magnets release, and gravity drops the rods into the core (a fail-safe SCRAM).

10 Control Strategy 2: Chemical Shim and Burnable Poisons

Mechanical rods warp the axial flux profile. Therefore, in a commercial PWR, mechanical rods are mostly withdrawn during normal operation. They are used for startup, shutdown, and safety transients—**not** for slow burnup compensation over the 18-month cycle.

10.1 Soluble Boron (The ChemE Approach)

To hold down the massive excess reactivity of a fresh core evenly, we dissolve **Boric Acid (**H_3BO_3**)** directly into the primary coolant. As the U-235 burns up over 18 months, the operators slowly dilute the boron concentration to maintain criticality.

- **Boron Letdown Curve:** Starts at ≈ 1200 ppm (Start of Cycle) and is diluted to ≈ 10 ppm (End of Cycle).

10.2 The Safety Constraint: Moderator Temperature Coefficient

Chemical Shim introduces a severe physics constraint:

- Normally, if water heats up and expands, its density drops. This reduces moderation, which safely lowers reactor power.
- However, if the water is heavily poisoned with Boric Acid, water expansion also decreases Boron *out* density. Removing poison *increases* reactivity!
- **The Limit:** To prevent the reactor from having a dangerous positive temperature coefficient, the NRC strictly limits the maximum allowable soluble boron concentration.

10.3 The Solution: Burnable Poisons

If we want to run a reactor for 24 months (requiring highly enriched fresh fuel), the required soluble boron would exceed the NRC limit. To fix this, engineers physically coat a thin layer of neutron poison (like Gadolinium, Erbium, or Boron-10) directly onto the UO_2 pellets of specific rods.

- These are called **Integral Fuel Burnable Absorbers (IFBAs)**.
- They hold down the excess reactivity at the beginning of the cycle, allowing operators to use safe, low levels of soluble boron. As the cycle progresses, the thin IFBA coating quickly burns away, perfectly matching the depletion of the Uranium!

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